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Paleomagnetic study of the eastern sector of Chapala Lake and implications for the tectonics of west-central Mexico

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Abstract

Western Mexico is characterized by several large-scale tectonic depressions which have been interpreted in terms of active continental rifting, tectonic transpression and coastal sinistral lateral transport of terranes. In this paper we report results of a paleomagnetic study of 148 samples from 22 sites in the Neogene volcanics from the eastern sector of the Chapala graben (western end of the E–W Chapala-Tula fault zone). Characteristic remanent magnetization directions have been isolated after detailed thermal demagnetization for sixteen sites. Six sites present reverse polarities and ten sites present normal polarities. Two sites show a low-latitude VGP and are considered as transitional. The normal and reverse polarity directions are almost antipodal. The overall characteristic pole position for the normal and reverse polarity sites, $N = 16$, $P_{LAT} = 74°N$, $P_{LONG} = 160°E$, $K = 25$ and $A_{95} = 7.6°$, lies to the left of the Neogene segment of the North American or northern Mexico apparent polar wander path, which suggest the occurrence of a counterclockwise rotation of -15.5 ± 7.4 to $-16.5 \pm 6.9°$. Results are interpreted in terms of counterclockwise vertical-axis rotation associated with regional transtension and left-lateral shear as a result of oblique subduction of the Cocos plate along the Middle American trench. We suggest that the Chapala graben developed early in the Miocene, in a left-lateral strike-slip environment within the regional E–W Chapala–Tula fault zone. Results support recent studies that propose relative motion for southern Mexico along the volcanic arc and trench-parallel strike-slip faulting.

1. Introduction

The present plate tectonic configuration of several small oceanic plates—Juan de Fuca, Gorda, Rivera, Cocos and Nazca—has long been interpreted as a result of Cenozoic plate interactions and break-up of the Farallon plate along the western margin of the North American plate (Atwater, 1970, 1989). The spatial and temporal characteristics of arc magmatism and crustal deformation have been investigated in an attempt to understand the kinematics and dynamics of plate tectonic evolution.

For western Mexico, this has not been an easy task and even the present-day plate tectonic boundaries and motions are not fully documented (Minster and Jordan, 1979; Klitgord and Mammerickx, 1982; Eissler and McNally, 1984; Johnson and Harrison, 1990; Bandy, 1992). Luhr et al.

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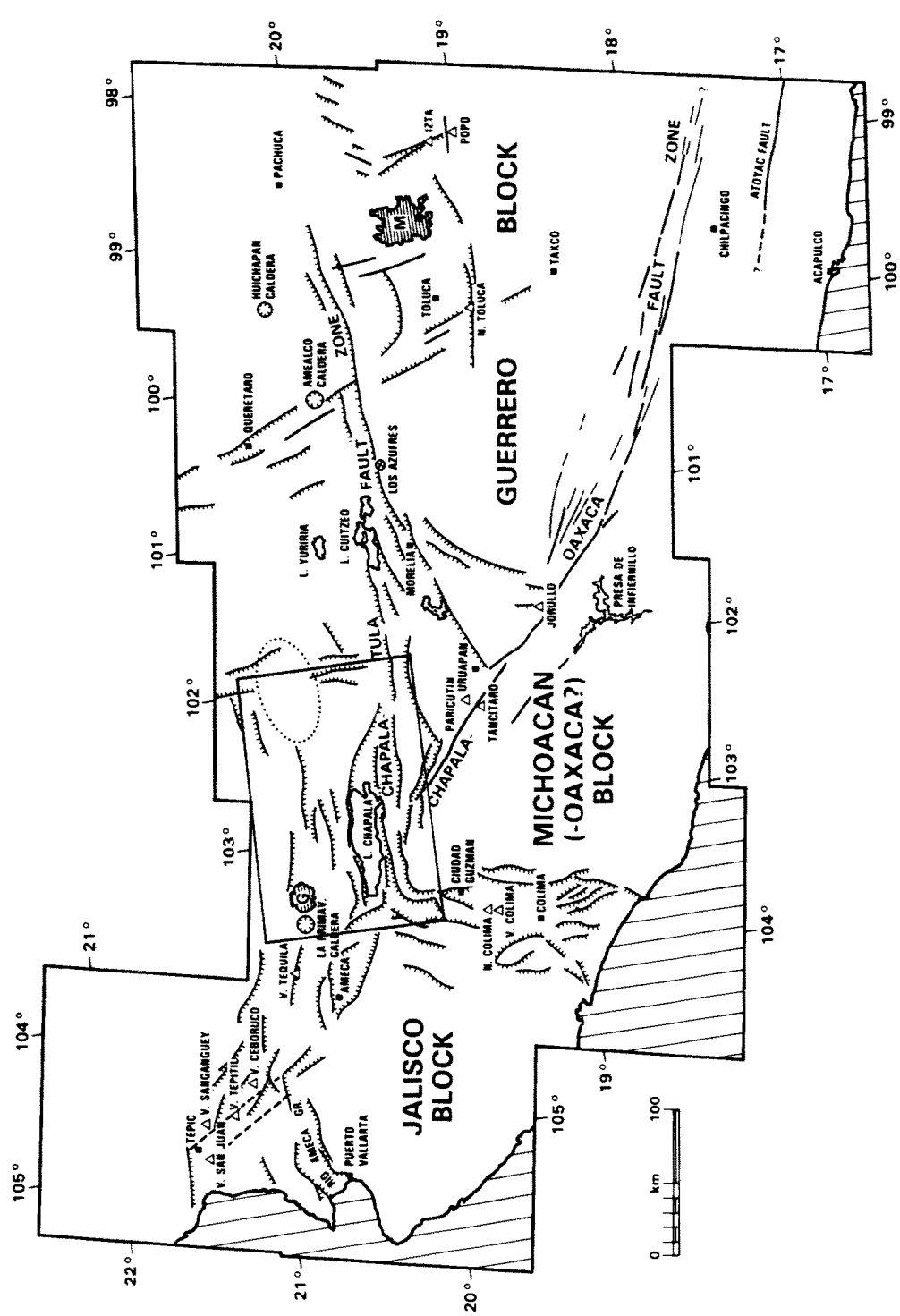


Fig. 1. Schematic simplified neotectonic map for central-western Mexico (taken from Johnson and Harrison, 1990) showing the study area (rectangle, see Fig. 2) in the western sector of the Chapala–Tula fault zone, within the Chapala graben. Active volcanoes are indicated by triangles; faults and lineaments are interpreted from Landsat TM images and field studies. Proposed major structures, i.e., Tepic–Chapala graben, Colima graben, Chapala graben, Chapala–Tula fault and Chapala–Oaxaca fault, are used to separate central-western Mexico into three tectonic blocks, Jalisco, Michoacan–Oaxaca and Guerrero blocks.

(1985) proposed the occurrence of active continental rifting in southwestern Mexico and the presence of an incipient rift–rift–rift triple junction formed by a set of prominent grabens: Colima, Tepic and Chapala (Fig. 1). The Colima and Tepic grabens bound the Jalisco block which in this interpretation is in the process of being rifted away from mainland Mexico and transferred to the Pacific plate (Allan et al., 1992). Active rifting is restricted to the Colima and Tepic grabens whereas the Chapala graben will then constitute an aulacogen.

Alternatively, modeling of present-day plate motions in the region suggests the presence of a sinistral component of motion in the order of 0–10 mm/yr (Johnson and Harrison, 1990; DeMets and Stein, 1990). This is interpreted in terms of southeastward coastal transport of southern Mexico, which implies that the Colima graben is a passive pull-apart zone instead of an active incipient rift. The Colima graben has also been interpreted in terms of an aulacogen (Nixon et al., 1986) where active extension vanished some time ago. The Chapala graben has been interpreted as a pull-apart basin related to a regional left-lateral strike-slip fault system (Urrutia-Fucugauchi, 1984, 1989). Johnson and Harrison (1990) have identified the presence of two major fault zones: the Chapala–Tula zone and the Chapala–Oaxaca zone (Fig. 1).

Several studies in the past have investigated the regional crustal deformation and strike-slip faulting in central and southern Mexico, particularly along the trace of the active magmatic arc (Molnar and Sykes, 1969; Mooser, 1972; Urrutia-Fucugauchi and Del Castillo, 1977). Relative differential motion among distinct portions of Mexico and the Rivera, Cocos and Pacific plates has been difficult to document, thereby complicating the development of a kinematic model (Johnson and Harrison, 1990). Recently a variety of studies involving field mapping, structural surveys, analysis of Landsat Thematic Mapper (TM) images, stress state, paleomagnetism and seismologic studies are improving the data base and permitting development and testing of kinematic models (e.g., Urrutia-Fucugauchi and Böhnel, 1988; Pasquaré et al., 1987, 1988; Johnson and Harrison, 1990; Nieto-Obregon et al., 1992; Suter et al., 1992).

In this paper we report on a paleomagnetic study of Neogene volcanics from the eastern sector of the Chapala graben (Fig. 1) and discuss its tectonic implications for west-central Mexico. This work follows a geological recconnaissance survey and stratigraphic and K-Ar dating studies in the area (Delgado-Granados and Urrutia-Fucugauchi, 1986; Rosas-Elguera et al., 1989). The paleomagnetic work forms part of a long-term project to study the tectonics and stratigraphy of the continental volcanic arc of central and southern Mexico. Arc magmatism occurs within an E–W-elongated province, the Trans-Mexican volcanic belt (TMVB) which presents a peculiar oblique orientation with respect to the Middle American trench. Available paleomagnetic data for the TMVB are briefly summarized and discussed in terms of their regional tectonic (geometric and kinematic) implications.

2. Geological setting and sampling

The Chapala graben is some 115 km long and about 37 km wide and is mostly occupied by the Chapala Lake (the largest in Mexico). The paleomagnetic study concentrated in the eastern sector of the graben, in the volcanic units that border the lacustrine plain of Chapala. The plain represents approximately one third of the graben and is bounded by a set of faults, of which the Ixtlan–Encinal, Pajacuaran and Nogales faults are the largest (Fig. 2). The Ixtlan–Encinal fault presents a curved trace in a NW–SE direction and can be followed for some 35 km. The fault affects the Encinal volcano; a K-Ar date of 3.3 ± 0.3 Ma for a lava flow from the volcano (Rosas-Elguera et al., 1989) supports recent activity in the area. The Pajacuaran fault cuts the andesitic volcanoes of the Pajacuaran range and has a surface extension of about 18 km. The Nogales fault trends northeast–southwest, approximately normal to the Ixtlan–Encinal fault. It apparently offsets the surficial distribution of thermal manifestations which are absent to the east-southeast of the

intersection between the two faults (Rosas-Elguera et al., 1989).

Major normal faults conform a broad zone of deformation in the Chapala graben. The Ixtlan–Encinal fault extends west along the northern shore of Chapala Lake (Mooser, 1972). The area extending some 50 km to the north of the lake is extensively faulted, with blocks 10–15 km in length and some 2–4 km wide and which dip gently to the north. Johnson and Harrison (1990) estimate fault offsets from the TM images of up to 1 km. The area to the south of the lake is also characterized by fault blocks that tilt slightly to the south. The major fault corresponds to the Pajacuaran fault that here presents maximum normal offsets of about 800 m (Johnson and Harrison, 1990). Spectral analysis of an aeromagnetic NNW–SSE profile across the central part of the lake shows normal offsets for the main graben in the order of 1 km, with the offset slighly smaller (some 200 m) along the southern fault than in the north (Campos-Enriquez et al., 1990).

Sampling sites for the paleomagnetic study were selected from the field survey and cover the age range of volcanic activity in the region (Rosas-Elguera et al., 1989). The paleomagnetic polarities also help to further constraint the age divisions. The radiometric dates span the range

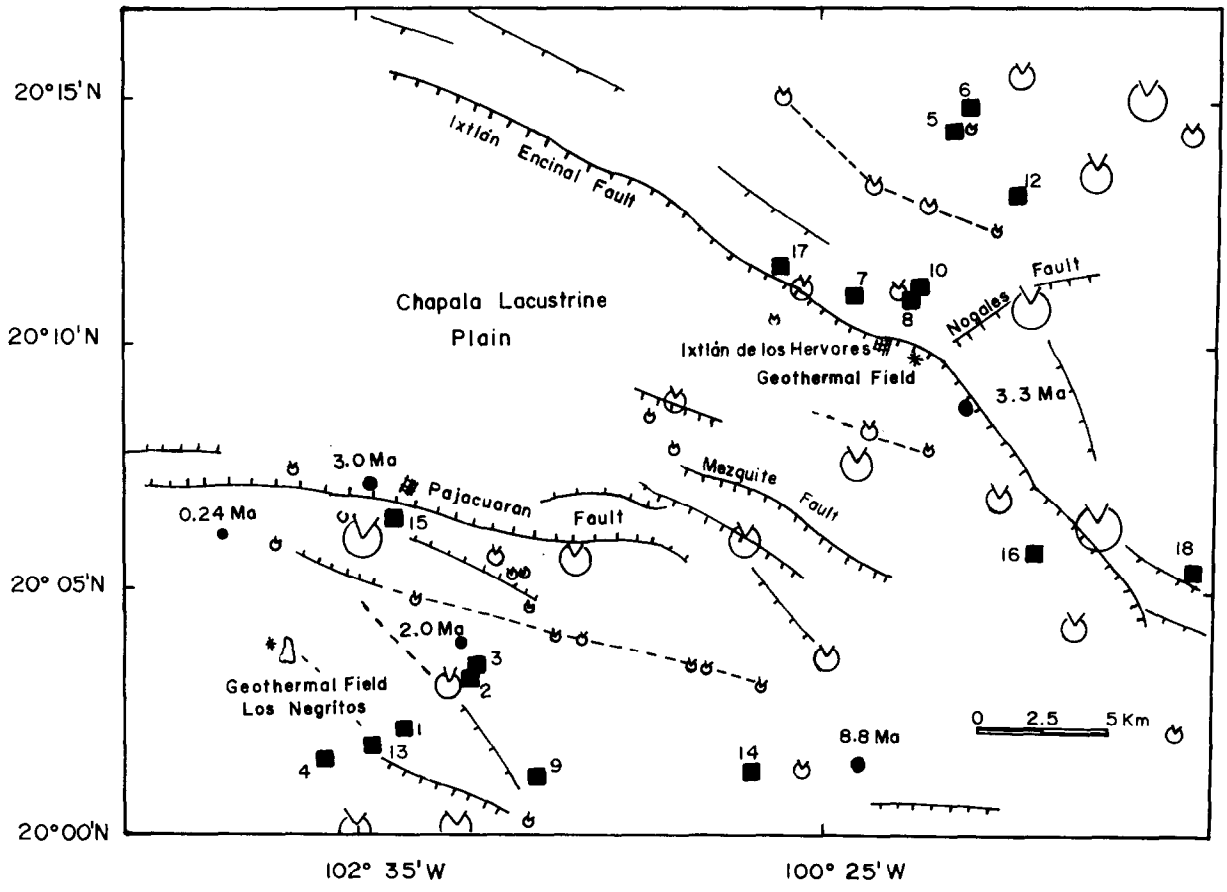


Fig. 2. Location of sampling sites for paleomagnetism (squares: numbers indicate site identification, see Table 1) and units dated by K-Ar (dots: dates in Ma). Major faults and volcanic structures are also indicated. The circles with the segments cut out of their tops represent volcanic craters of the structures; size of the circles is a relative measure of the size of the volcanoes. The discontinous lines represent inferred faults. Mapped faults are shown by the continous lines; down-faulted block is indicated by the small line pattern. Note that faulting affects some of the volcanic structures (e.g., Nogales fault and Ixtlan–Encinal fault).

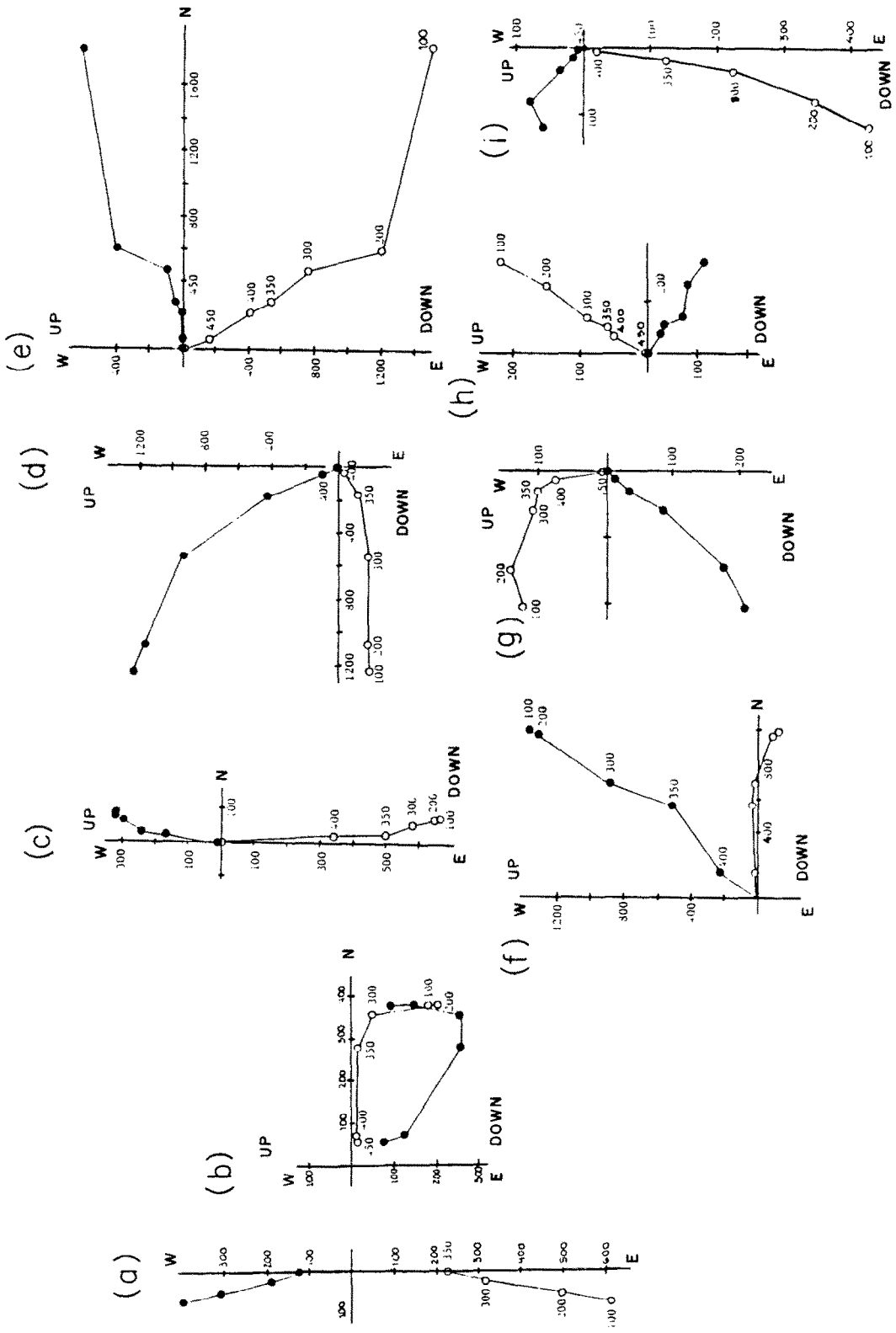


Fig. 3. Examples of orthogonal Zijderveld plots (thermal demagnetization; values correspond to temperature steps) for units studied.

from 9 to 0.24 Ma (Fig. 2). The magnetic polarities and K-Ar dates indicate activity mainly within the Matuyama and Gauss chrons.

Chemical composition of the volcanic products in the area has changed through time from subalkaline in the Miocene to alkaline since the Pliocene (Rosas-Elguera et al., 1989).

A total of 148 cores was drilled at 22 sites with a petrol-powered drill and oriented in situ with a Brunton magnetic compass. Orientations were corrected by the present-day magnetic declination for the area which is N8°E. Location of paleomagnetic sampling sites and K-Ar dated samples is included in Fig. 2 (mean coordinates for the area are 20.16°N and 102.5°W).

3. Paleomagnetic results

The direction and intensity of natural remanent magnetization (NRM) were measured with a Molspin fluxgate magnetometer. Both normal and reverse polarities are present. The vectorial composition and stability were investigated by detailed step-wise thermal demagnetization. Eight to ten steps up to 600°C were used with a non-inductive Schonstedt TSD-1 furnance. Vector substraction, principal component and end-point analyses (Zijderveld, 1967; Kirschvink, 1980) were employed to estimate the vector components. Examples of the different behaviours observed during demagnetization are included in Fig. 3. Most

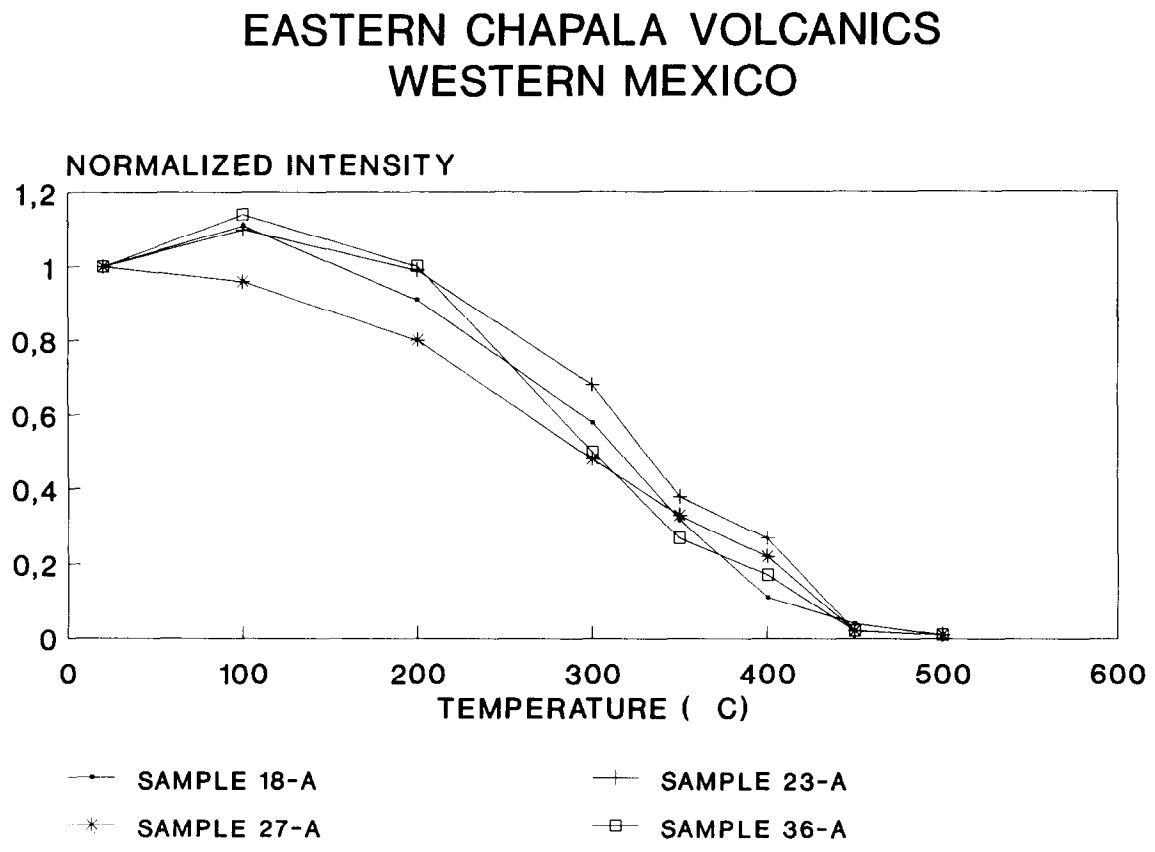


Fig. 4. Normalized intensity plots for some units studied. Note that maximum unblocking temperatures do not exceed 400–450°C.

samples were characterized by two components with discrete or overlapping spectra. A characteristic magnetization corresponding to a linear component going through the origin in the vector plots was isolated in most samples. Maximum unblocking temperatures are lower than 450°C, suggesting that magnetic carriers belong to the titanomagnetite series (Fig. 4).

Vector means and associated Fisher statistics were calculated for each site. Four sites gave scattered directions (i.e., $k < 10$ and $\alpha_{95} > 25$) and are not further considered for discussion. Results for sixteen sites are summarized in Table 1 and illustrated in Fig. 5. Ten sites are of normal polarity and six sites are of reverse polarity. Virtual geomagnetic pole (VGP) pole positions were calculated from the corresponding site mean directions. Two additional sites show low-latitude VGPs and are considered of intermediate or transitional polarity. Mean directions and pole positions for the normal polarity group and the reverse polarity group are included in Table 1. The two mean directions are almost antipodal, within the statistical uncertainties. The overall mean direction and pole position for the eastern Chapala volcanics (calculated without the two sites with low-latitude VGPs) are also given in Table 1.

Age assignments in Table 1 are derived from the field studies and K-Ar dating and referred to magnetic polarity chrons (Rosas-Elguera et al., 1989). The magnetostratigraphic Time Scale used is from A. Cox (in Harland et al., 1990). Units correlated to the Matuyama chron are of reverse polarity (sites 5, 6 and 7), intermediate (site 8) and normal polarity (sites 2 and 3). Site 3 in the

Table 1
Summary of paleomagnetic data for the eastern Chapala volcanics

No.	Site	N	Dec	Inc	k	α95	Lat(N)	Long(E)	Polarity (K-Ar)
1	Basura	10	346.1	28.4	451	2.3	75.9	149.3	N
2	Moral	8	346.2	31.5	231	3.6	76.6	156.8	N
3	Varal	7	343.4	24.6	47	8.9	72.5	145.6	N 2.0 ± 0.2
4	Villamar	12	350.3	28.8	54	6.0	79.6	142.4	N
5	Gertrudiz	8	180.9	−41.3	21	12.4	86.4	90.8	R
6	Pelon	8	157.5	−39.6	29	10.4	68.9	182.3	R
7	Coruco	7	135.0	−34.1	27	11.8	47.6	186.4	R
8	Trompeta	6	114.1	−15.6	12	20.3	25.3	193.0	I
9	Zarquillo	8	251.0	9.4	14	15.4	16.0	181.2	I
10	Sn.Fco.	8	326.6	37.8	37	11.0	58.8	175.5	N
11	Venustian	6	333.0	5.4	18	16.4	59.0	137.3	N
12	Bolita	7	12.9	33.8	25	12.3	77.7	353.4	N
13	Salitre	7	166.1	−13.7	50	8.6	71.2	125.1	R
14	Quirosena	10	146.5	−36.1	81	5.4	58.6	186.5	R 8.8 ± 0.8
15	Muerto	6	343.9	33.6	273	4.1	74.7	165.8	N 3.0 ± 0.3
16	Sauceda	6	158.8	−27.7	89	7.1	69.1	158.0	R
17	Puente	6	334.1	25.5	118	6.2	64.3	158.5	N
18	T.Blancas	5	350.2	36.1	207	5.3	80.8	170.8	N
Reverse polarity group									
		6	157.4	−32.9	24	13.9	68.5	167.6	
Normal polarity group									
		10	344.4	29.1	33	8.6	74.5	155.1	
Overall mean									
		16	343.5	30.8	25	7.6	74.0	159.7	

No., number for site identification; N, number of samples; Dec, Inc, declination and inclination of characteristic magnetization; k, α_{95}, confidence parameter and 95% cone of confidence for Fisher statistics; Lat, Long, latitude and longitude for VGP; Polarity is N (normal), I (intermediate) and R (reverse). K-Ar dates in Ma ($\pm 1\ \sigma$). Overall mean direction and pole calculated with all sites, except sites 8 and 9.

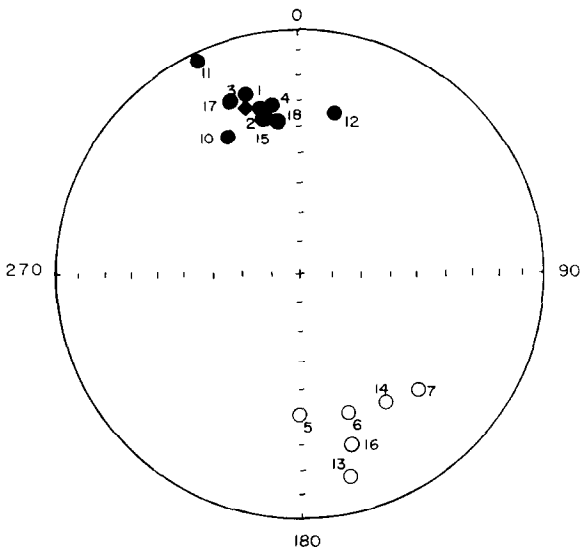


Fig. 5. Site mean directions plotted in a stereographic equal-area projection (see Table 1). Both normal and reverse polarities are present. Note the good overall agreement of mean directions which are roughly antipodal. Sites 8 and 9 which show low-latitude VGPs are not plotted. Overall mean direction is indicated by the rhomb.

Varal volcano has been dated at 2.0 ± 0.2 Ma and may then correlate with the Reunion normal subchrons of 2.04–2.01 and 2.14 to 2.12 Ma. The site mean directions from sites 2 (Moral) and site 3 are similar; the A_{95} cones of confidence overlap, suggesting they come from the same population. The test described by McFadden and Lowes (1981) corresponding to the case when the concentration parameters k_1 and k_2 are different, has been applied. The following statistics is used:

$$\frac{r\left[(R_1+R_2)^2-R^2\right]}{2\left[(N_1-R_1)+r(N_2-R_2)\right](R_1+rR_2)}=A$$

$$A>\frac{1}{p}^{1/(N-2)}-1$$

where N_i is the number of samples per site, R_i is the site vector length, $r=k_2/k_1$, with $k_2>k_1$, p is the significance level (0.05) and $N=N_1+N_2$.

The two directions are not independent at the 95% confidence level (we obtain the values of $0.1719>0.2592$ for the expression above). The

Table 2
Summary of paleomagnetic data for the Trans-Mexican volcanic belt

Locality	N	Lat (°N)	Long (°E)	A_{95}	$R\pm\Delta R$	$F\pm\Delta F$	Age
1. E. Mexico volc.	28	75.1	170.7	5.1	−15 ± 6	−1 ± 6	Q(B)
2. E. Mexico volc.	20	71.6	166.5	7.4	−19 ± 7	2 ± 7	M–P
3. Basin Mexico	42	87.5	164.5	3.0	−2 ± 4	−3 ± 6	Q(B)
4. Basin Mexico	19	88.1	302.8	6.5	5 ± 7	−4 ± 6	eM–P
5. Basin Mexico	22	88.4	79.3	7.2	8 ± 8	6 ± 8	eO–M
6. Jantetelco–Tepexco	9	33.7	176.3	10.0	−56 ± 13	6 ± 13	M
7. NE Jalisco	7	68.1	181.1	(10.0)	−20 ± 14	1 ± 12	O–M
8. Guerrero volc.	6	54.8	164.5	8.6)	−32 ± 11	11 ± 12	O–M
9. Balsas Fm.	14	54.1	183.4	(12.0)	−40 ± 14	18 ± 14	Pa–E
10. E. Mexico L.	4	32.6	195.2	20.2	−41 ± 26	−6 ± 27	mK
11. Acambay volc.	17	72.5	170.3	(7.5)	−16 ± 7	3 ± 8	P–Q
12. R. Santiago	3	81.2	128.1		−6	6	P
13. R. Santiago	4	79.1	180.6		11	−6	M
14. Amatlan volc.	8	81.9	197.2	(9.1)	−8		Q
15. E. Chapala L.	16	74.0	159.7	7.6	−17 ± 7	3 ± 6	M–Q

N is the numer of sites; $R\pm\Delta R$ is the rotation parameter and 95% statistical uncertainty; $F\pm\Delta F$ is flattening parameter and 95% statistical uncertainty (Beck, 1980; Demarest, 1983). Age: Q = Quaternary; Q(B) = Quaternary (Brunhes chron); P = Pliocene; M = Miocene; eM = early Miocene; O = Oligocene; eO = early Oligocene; E = Eocene; Pa = Paleocene; mK = mid-Cretaceous. Entries 1 to 10 are taken from the compilation by Urrutia-Fucugauchi and Bohnel (1988); entry 11 is from Soler-Arechalde and Urrutia-Fucugauchi (1994); entries 12 and 13 are calculated from data given in Watkins et al. (1971); entry 14 is calculated from data given in Nieto-Obregon et al. (1992); and entry 15 is from this paper.

K-Ar date for site 3 (Varal) is 2.0 ± 0.2 Ma (Table 1). Therefore, on the basis of just this information the two sites are assumed to correlate with one of the Reunion subchrons.

The two other sites with reverse polarity are correlated with the Gilbert chron and chron number 8 (this with a K-Ar date of 8.8 ± 0.8 Ma). The other sites with normal polarity correspond to the Brunhes chron (sites 1 and 4) and to the Gauss chron (sites 10, 11 and 12). Sites 8 (Trompeta) and 9 (Zarquillo) with a low-latitude VGP require further study before referring it to the polarity time scale.

4. Discussion

For the analysis, we calculated the mean pole for all sites at 74.0°N and 159.7°E, with $A_{95} = 7.6$ and $K = 24.5$. Most site mean directions are rotated counterclockwise from the dipolar direction for the area and there is no apparent differences according to age in the distribution of remanence directions. The normal and reverse directions are antipolar. Assuming that the mean direction and pole estimated for the sixteen sites (Table 1) are representative for the volcanic units, they can be compared with the expected direction and pole for the Neogene. The mean pole lies to the left of the reference poles estimated for cratonic North America (Irving and Irving, 1982) and northern Mexico (Urrutia-Fucugauchi, 1984). The angular difference may be quantified by using the rotation (R) and flattening (F) parameters (Beck, 1980) and their statistical uncertainties (Demarest, 1983). This results in a R ($\pm\Delta R$) of $-15.5°$ (± 7.4) to $-16.5°$ (± 6.9) and a F ($\pm\Delta F$) of 2.7° (± 6.3) to 5.5° (± 6.0) [reference pole for 10 Ma from Irving and Irving (1982) and the geographic pole]. The negative R value is similar to results from other areas in central and eastern Mexico (Table 2) which suggests the occurrence of a counterclockwise rotation of the study area.

We suggest that the Chapala graben developed early in the Miocene, associated with left-lateral movements along an E–W fault system. The block rotations observed in the paleomagnetic data may be associated with formation of the structure as a large pull-apart basin. Tectonic activity has apparently migrated southwards and is mostly concentrated along the Pajacuaran fault and in the Citlala graben (Rosas-Elguera et al., 1989; Johnson and Harrison, 1990). Major tectonic activity in the Chapala graben is disappearing and correlates with a decline in volcanic activity. This is in contrast with the active tectonism and volcanism observed in the Colima and Tepic grabens (Allan et al., 1992). The trend of the Chapala–Tula fault zone is oblique to the trend of the trench and plate convergence direction (Fig. 1). Activity in the magmatic arc is associated with this E–W orientation which points to the dominant structural control of magmatism and the deformation of the crust. Estimates of the paleo-stress state since the Miocene indicate major changes with time as a result of the re-organization of plate interactions (e.g., Pasquaré et al., 1988; Suter et al., 1992). The trend of the Chapala–Oaxaca fault zone (Fig. 1) tends to be roughly parallel with the trench trend and it is tempting to conclude that this fault is presently taking part or most of the left-lateral displacements related to oblique subduction of the Cocos plate along the Middle America trench (e.g., DeMets and Stein, 1990). However, the historic shallow crustal seismic activity within the Acambay segment of the Chapala–Tula fault zone that indicates dominant normal faulting with a left-lateral component, the structural evidence of recent deformation, recent volcanic activity and the documentation of tectonic block rotations (Suter et al., 1992; Soler-Arechalde and Urrutia-Fucugauchi, 1994), argues for continuing tectonic activity within the Chapala–Tula fault and E–W crustal deformation in central Mexico.

The Mexican continental margin is subject to large-scale sinistral shear driven by oblique plate subduction along the trench and as a result the region is fragmented into large crustal blocks that experience counterclockwise tectonic rotations. This interpretation supports southeastward left-lateral transport of southern Mexico. Further work is required to investigate relative motion between the Michoacan and Guerrero blocks. The lateral extent of the two Chapala fault zones is well in excess of 300–400 km long (Fig. 1). The

Chapala–Oaxaca fault gradually becomes parallel to the trench trend to the south, whereas the Chapala–Tula fault remains oblique to the trench trend. The two correspond to major zones of crustal deformation and the Chapala–Tula fault exibits a major structural control for the magmatism in the volcanic arc.

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References

Allan, J.F., Nelson, S.A., Luhr, J.F., Carmichael, I.S.E., Wopat, M. and Wallace, P.J., 1992. Pliocene–Holocene rifting and associated volcanism in southwest Mexico: An exotic terrane in the making. In: The Gulf and Peninsular Province of the Californias. Am. Assoc. Pet. Geol. Mem. Ser., 47, Chap. 21.

Atwater, T., 1970. Implications of plate tectonics for the evolution of western North America. Geol. Soc. Am. Bull., 81: 3513–3536.

Atwater, T., 1989. Plate tectonic history of the northeast Pacific and western North America. In: E.L. Winterer et al. (Editors), The Eastern Pacific Ocean and Hawaii (The Geology of North America, N). Geol. Soc. Am., Boulder, CO, pp. 21–72.

Bandy, W., 1992. Geological and geophysical investigation of the Rivera–Cocos plate boundary: Implications for plate fragmentation. Ph.D. Thesis, Texas A and M Univ., College Station, TX.

Beck Jr., M.E., 1980. Paleomagnetic record of plate-margin tectonic processes along the western edge of North America. J. Geophys. Res., 85: 6153–6170.

Campos-Enriquez, J.O., Arroyo-Esquivel, M.A. and Urrutia-Fucugauchi, J., 1990. Basement, Curie isotherm and shallow crustal structure of the Trans-Mexican volcanic belt from aeromagnetic data. Tectonophysics, 172: 77–90.

Delgado-Granados, H. and Urrutia-Fucugauchi, J., 1986. Estructura del graben de Chapala. Com. Tec., Inst. Geofis., UNAM, 12: 1–20.

Demarest Jr., H.H., 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data. J. Geophys. Res., 88: 4321–4328.

DeMets, C. and Stein, S., 1990. Present-day kinematics of the Rivera plate and implications for tectonics in southwestern Mexico. J. Geophys. Res., 95: 21,931–21,948.

Eissler, H.K. and McNally, K.C., 1984. Seismicity and tectonics of the Rivera plate and implications for the 1932 Jalisco, Mexico earthquake. J. Geophys. Res., 89: 4520–4530.

Harland, W.B., Armstrong, R., Cox, A., Craig, L., Sith, A. and Smith, D., 1990. A Geological Time Scale 1989. Cambridge Univ. Press, New York, NY, 263 pp.

Irving, E. and Irving, G.A., 1982. Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana. Geophys. Surv., 5: 141–188.

Johnson, C.A. and Harrison, C.G.A., 1990. Neotectonics in central Mexico. Phys. Earth Planet. Inter., 64: 187–210.

Kirschvink, J., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J.R. Astron. Soc., 62: 699–718.

Klitgord, K.D. and Mammerickx, J., 1982. East Pacific rise: Magnetic anomaly and bathymetric framework. J. Geophys. Res., 87: 6725–6750.

Luhr, J.F., Nelson, S.A., Allan, J.F. and Carmichael, I.S.E., 1985. Active rifting in southwestern Mexico: Manifestations of an incipient eastward spreading-ridge jump. Geology, 13: 54–57.

McFadden P.L. and Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions. Geophys. J.R. Astron. Soc., 67: 19–33.

Minster, J.B. and Jordan, T.H., 1979. Rotation vectors for the Phillipine and Rivera plates. EOS, Trans. Am. Geophys. Union, 60: 958 (abstract).

Molnar, P. and Sykes, L.R., 1969. Tectonics of the Caribbean and Middle America regions from focal mechanisms and seismicity. Geol. Soc. Am. Bull., 80: 1639–1684.

Mooser, F., 1972. The Mexican volcanic belt. Structure and tectonics. Geofis. Int., 12: 55–70.

Nieto-Obregon, J., Urrutia-Fucugauchi, J., Cabral-Cano, E. and Guzman de la Campa, A., 1992. Listric faulting and continental rifting in eastern Mexico — A paleomagnetic and structural study. Tectonophysics, 208: 365–376.

Nixon, G.T., Demant, A., Armstrong, R.L. and Harakal, J.E., 1987. K-Ar and geologic data bearing on the age and evolution of the Trans-Mexican volcanic belt. Geofis. Int., 26: 109–158.

Pasquaré, G., Ferrari, L., Perazzoli, V., Tiberi, M. and Turchetti, F., 1987. Morphological and structural analysis of the central sector of the Transmexican volcanic belt. Geofis. Int., 26: 177–193.

Pasquaré, G., Garduño, V.H., Tibaldi, A. and Ferrari, M., 1988. Stress pattern evolution in the central sector of the Mexican volcanic belt. Tectonophysics, 146: 353–364.

Rosas-Elguera, J.G., Urrutia-Fucugauchi, J. and Maciel, R., 1989. Geologia del extremo oriental del graben de Chapala: Breve discusion sobre su edad — Zonas geotermicas Ixtlan de los Hervores–Los Negritos, Mexico. Geotermia, 5: 3–18.

Soler-Arechalde, A.H. and Urrutia-Fucugauchi, J., 1994. Regional rigid-block rotation, small domain rotations and distributed deformation within the Acambay graben, central Trans-Mexican volcanic belt: Paleomagnetic implications. Geofis. Int. 34: 565–574.

Suter, M., Quintero, O. and Johnson, C.A., 1992. Active faults and state of stress in the central part of the Trans-Mexican volcanic belt, Mexico 1. The Venta de Bravo fault. J. Geophys. Res., 97: 11,983–11,993.

Urrutia-Fucugauchi, J., 1981. Palaeomagnetism of the Miocene Jantetelco granodiorites and Tepexco volcanic group and inferences for crustal block rotations in central Mexico. Tectonophysics, 76: 149–168.

Urrutia-Fucugauchi, J., 1984. On the tectonic evolution of Mexico: Paleomagnetic constraints. In: R. Van der Voo et al. (Editors), Am. Geophys. Union, Geodyn. Ser., 12: 29–47.

Urrutia-Fucugauchi, J., 1989. Rotacion de bloques en sistemas de fallas de desplazamiento lateral-cuencas tensionales. Geofis. Int., 28: 907–938.

Urrutia-Fucugauchi, J. and Böhnel, H., 1988. Tectonics along the Trans-Mexican volcanic belt according to paleomagnetic data. Phys. Earth Planet. Inter., 52: 320–329.

Urrutia-Fucugauchi, J. and Del Castillo, L., 1977. Un modelo del eje volcanico mexicano. Bol. Soc. Geol. Mex., 38: 18–28.

Watkins, N.D., Gunn, B.M., Baksi, A.K., York, D. and Ade-Hall, J., 1971. Paleomagnetism, geochemistry and potassium-argon ages of the Rio Grande de Santiago volcanics, central Mexico. Geol. Soc. Am. Bull., 82: 1955–1968.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: Analysis of results. In: D.W. Collinson et al. (Editors), Methods in Palaeomagnetism. Elsevier, Amsterdam, pp. 254–286.